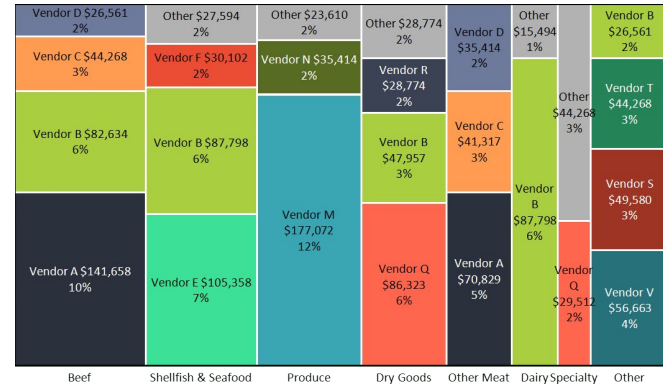
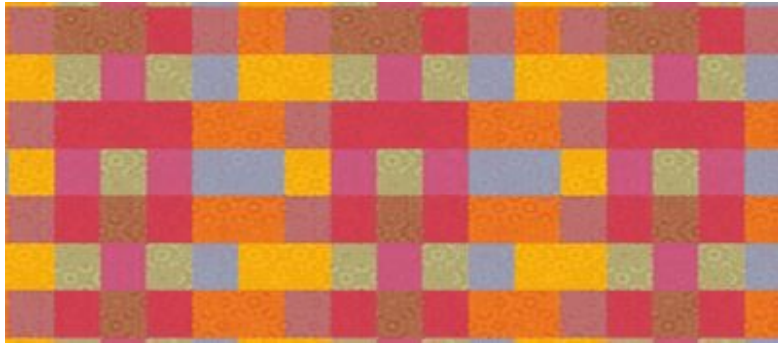


Marimekko Chart

Chaitanyasuma Jain

Inspiration behind Marimekko Charts

- The Marimekko design firm of Helsinki, Finland exemplified the exuberant and colorful design style of the 1960s. The firm's vivid fabrics and furnishings were a favorite among the decade's bold and beautiful.



What is a Marimekko Chart?

- A combination of a 100% stacked column chart and 100% stacked horizontal-bar chart using a different variable for each
- A variable-width stacked column chart
- A way to show part-to-whole relationships across two variables at once
- A way to show the frequencies of a contingency table where the area of each displayed cell is proportional to the whole

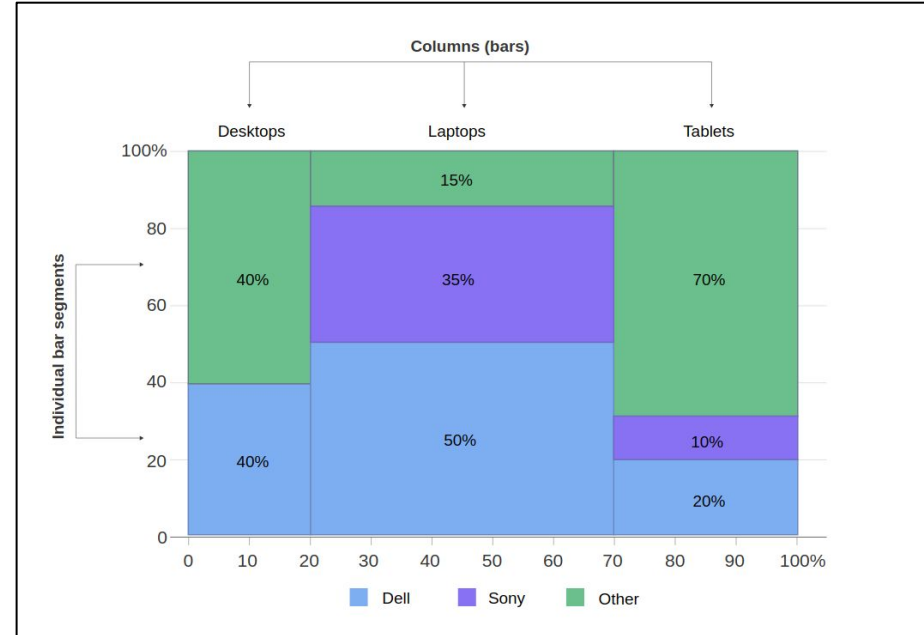
Derived from

Stacked Bar Graph



How to read the chart

- Columns (bars)
- Individual bar segments
- Colors



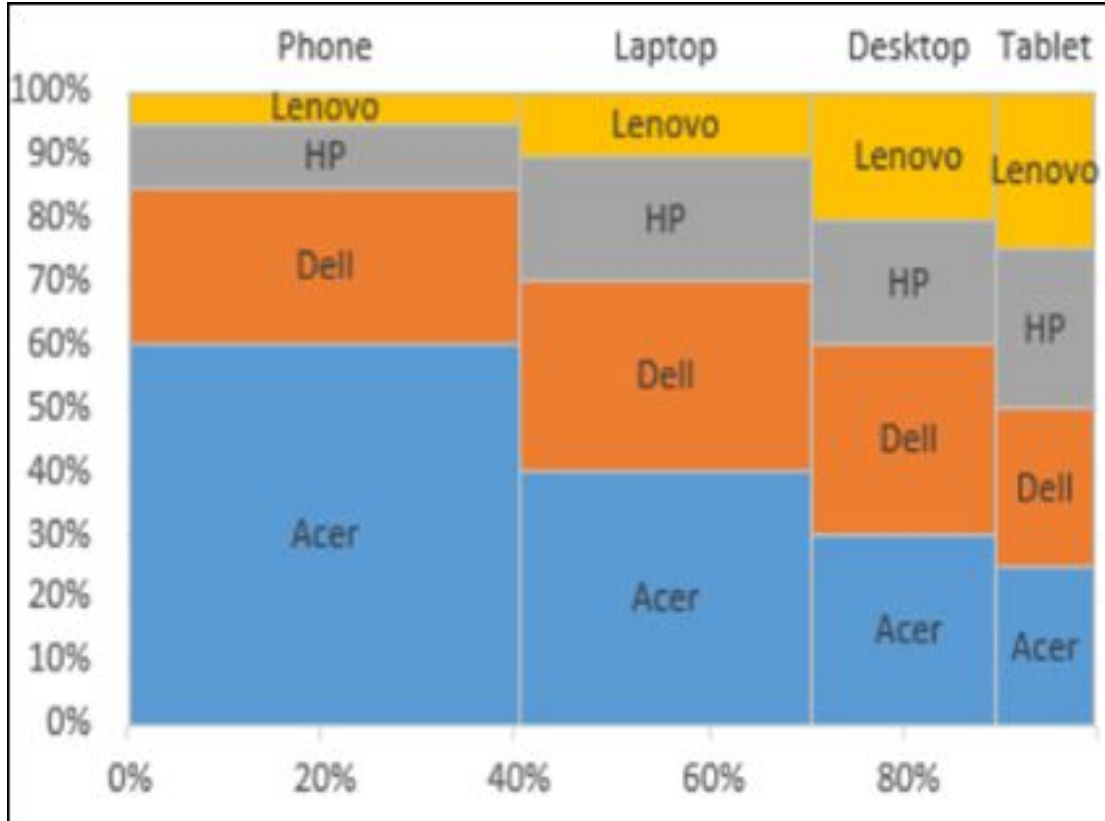
Alternative Names & How to draw on paper

Also called mosaic plot, matrix chart, stacked spinogram, spineplot, olympic or submarine chart, a Mondrian diagram, or even shortened to just mekko chart.

To draw on paper:

1. All heights of columns need to be equal. Thus, scale the width accordingly (first variable).
2. Within each column, represent the second variable by adjusting its heights.

Real-Life Applications - Case 1

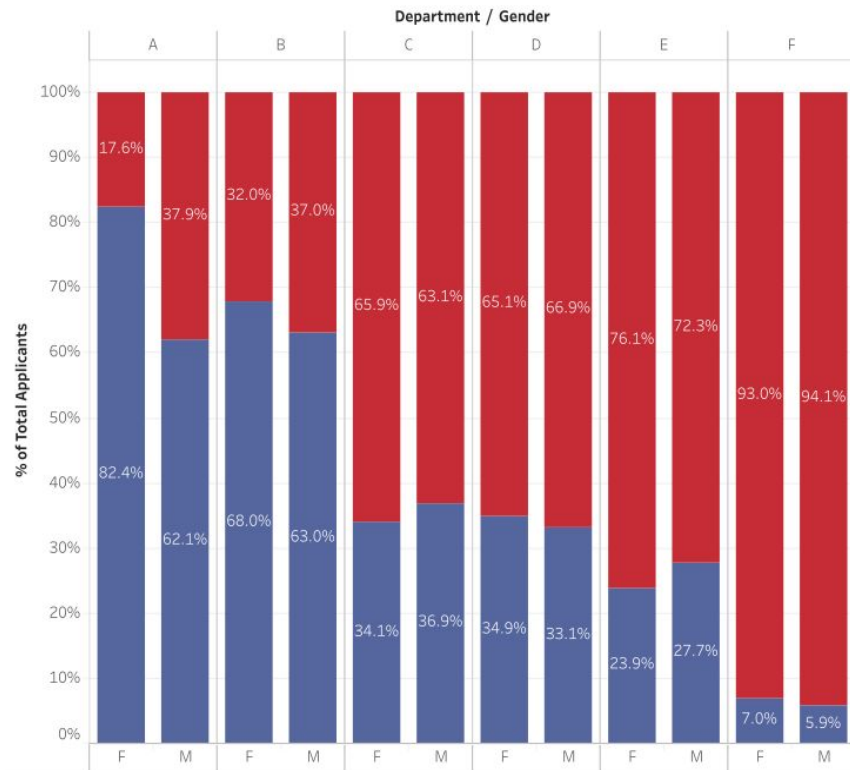


It is popular in business, particularly where marketing departments analyze segmentation of their markets. Market segments are often arrayed along the x-axis, with the width of each column corresponding to the dollar size of a segment.

The following three dimensions of marketing data can be represented using a Marimekko chart: Total value of a market segment, Combined share of all competitors in a market segment, Individual share of competitors in a market segment

Real-Life Applications – Case 2

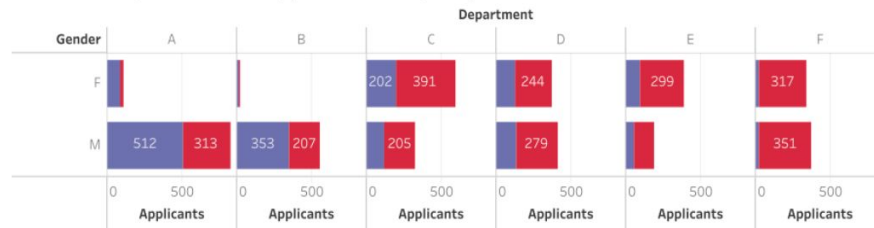
UC Berkeley Graduate Admissions Data for the Six Largest Departments



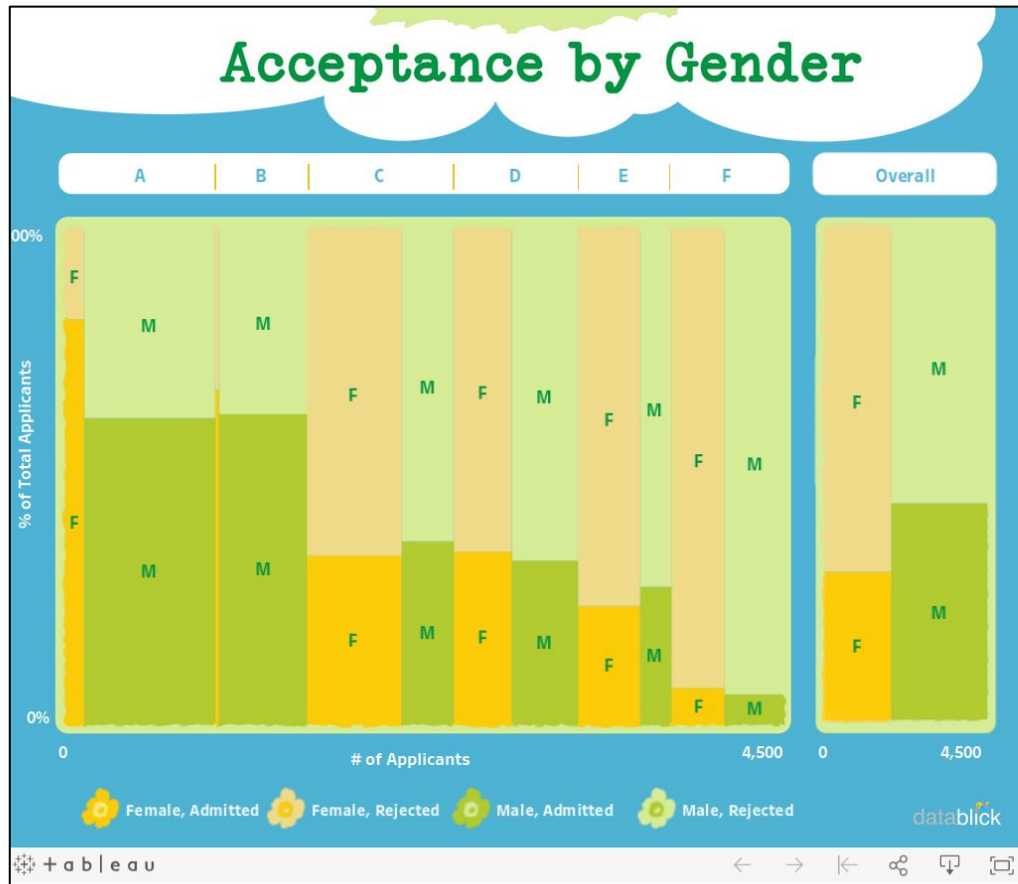
Admission Status

Rejected
Admitted

UC Berkeley Graduate Applications by Department



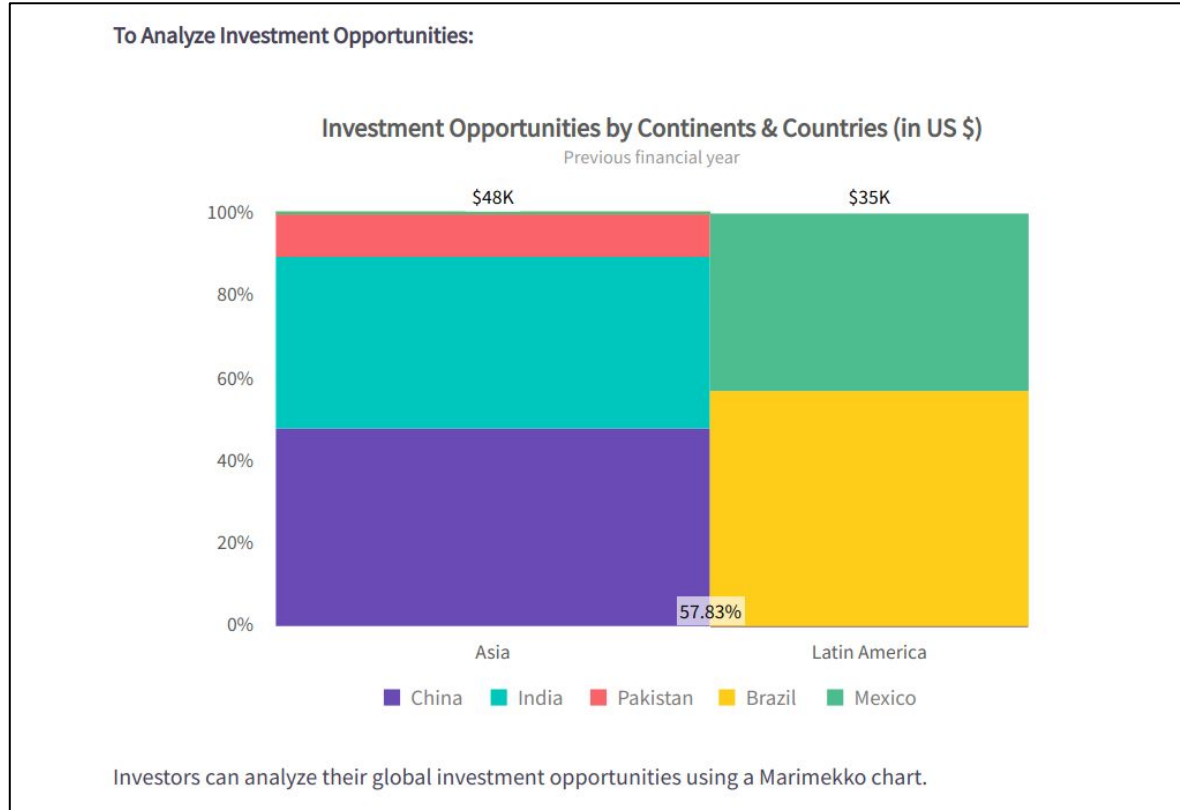
Real-Life Applications – Case 2



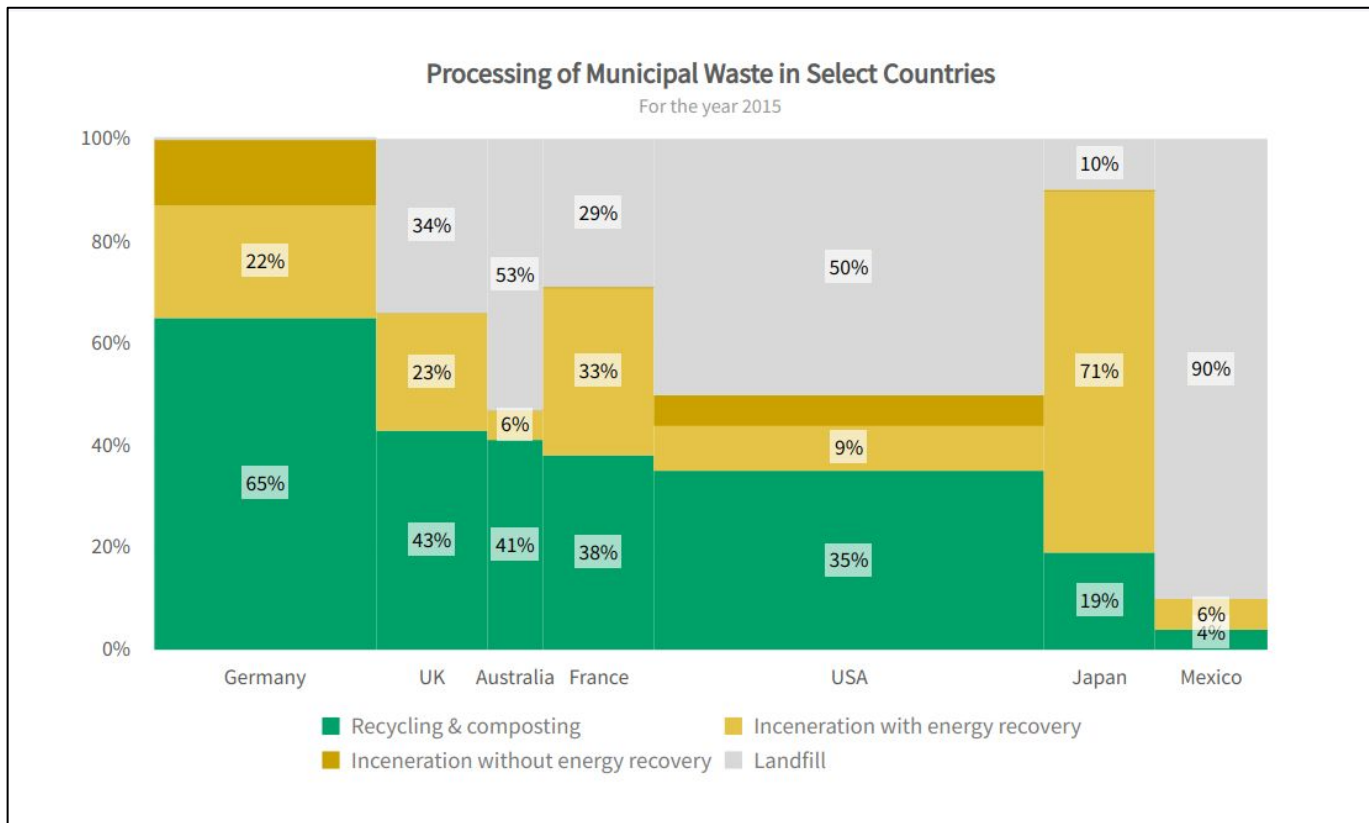
A group of UC Berkeley faculty and staff obtained detailed admissions data and used a marimekko chart to claim a bias against women when it came to admission applications.

Ref: <https://www.tableau.com/about/blog/2016/8/introduction-marimekko-chart-many-colors-and-many-names-58111>

Real-Life Applications – Case 3



Real-Life Applications – Case 4



More Real-Life Applications

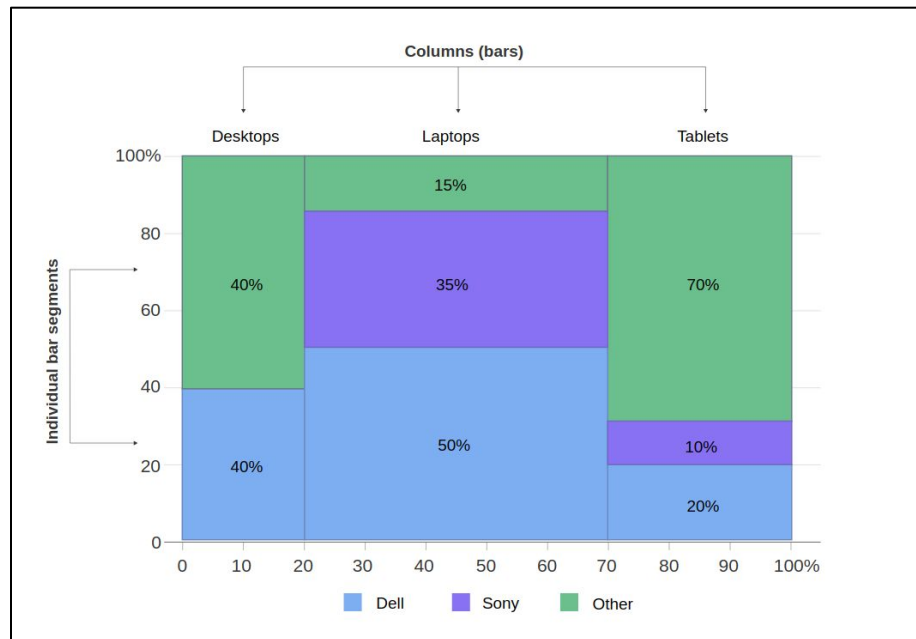
Find here - <https://www.mekkographics.com/product/power-of-mekko-charts/>,
[https://www.anychart.com/products/anychart/gallery/Mekko \(Marimekko\) and Mosaic Charts/](https://www.anychart.com/products/anychart/gallery/Mekko%20(Marimekko)%20and%20Mosaic%20Charts/)

Implementation

- JSON <https://www.fusioncharts.com/dev/chart-guide/standard-charts/marimekko-chart>
- D3 <https://observablehq.com/@d3/marimekko-chart>
- Available in Tableau
https://www.tableau.com/about/blog/2016/8/how-build-marimekko-chart-tableau-58153?_src=li_ftigniter&_widget=blog-widget&li_source=LI&li_medium=blog-widget
- Can be made using Excel without add-on
<https://www.contextures.com/excelmarimekkochart.html>
- Excel add-ons <https://peltiertech.com/Utility30/>

Functions

- Comparison - Shows the differences or similarities between values. (with an axis)
- Part-to-a-whole - Shows part (or parts) of a variable to its total. Often used to show how something is divided up.
- Proportions - Uses size or area to show differences or similarities between values or for parts to a whole.
- Relationships - Shows relationships and connections between the data or show correlations between two or more variables.



Variations

- Marimekko Chart - A variable width 100% stacked bar chart. Bars of same height
- Bar Mekko Chart - A variable width bar chart



Differences with similar charts

- Stacked Bar Graph - Column/Row width remains same. Thus we can measure only one variable
- Tree Map - Area is the only spatial dimension used to encode data in treemaps. In the case of Marimekko charts, however, separate values are encoded by the length and the width of each rectangle. As such, all of the variables in a Marimekko graph are spatially encoded.



Advantages & Disadvantages

Advantages:

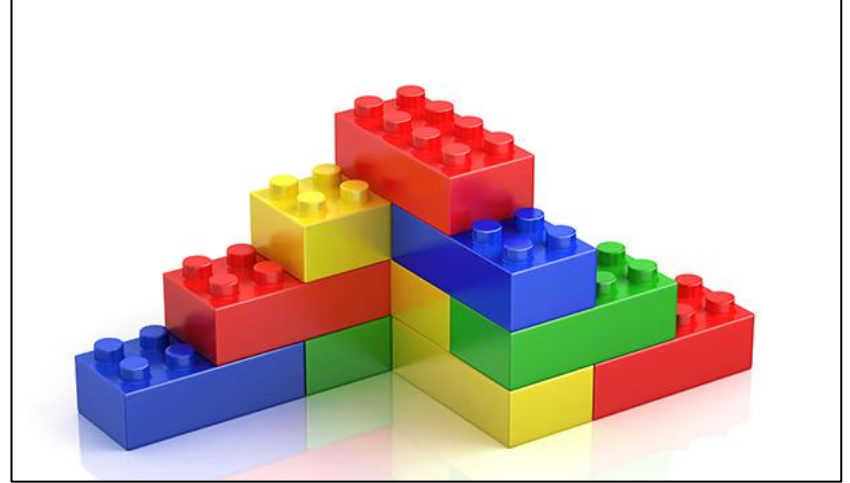
- Can compare two variables simultaneously.
- Visually appealing.
- Very popular for making presentations. Easy to make qualitative implications.

Disadvantages:

- It is criticised for making the data hard to perceive and to compare visually.
- In order to search for differences in only width or only height, we are required to work harder and spend more time than we would if focusing on the differences of two separable visual dimensions. This problem might not seem significant in the case of the graph above, but as the amount of data increases, it becomes more and more of a problem.
- Marimekko graphs suffer from a problem that plagues any stacked bar graph: It is difficult to accurately make comparisons of the width or height of boxes that are not arranged next to one another along a common baseline.

3D Modelling & Gamification

- Can be done using Lego blocks



Technical Paper

1. **The Great Eight: Trillion-Dollar Growth Trends to 2020** from Bain & Company used Marimekko Chart to answer 2 important questions related to their research – How much will the world's population grow between 2010 and 2020 and which countries will contribute most to the growth?
2. **Small data as a conversation starter for learning analytics: Exam results dashboard for first-year students in higher education** – Marimekko charts encoding the responses to the three statements – usefulness, clearness and reflection.